Assignment Title: Liquid Level Control using Siemens S7-1200 in TIA

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Submission date	
Name	
ID number	

What you will learn from this assignment?

- Writing SCL code for scaling the analog input
- Writing SCL code for Semi Automation ON/OFF Control
- Writing SCL code for Proportional Control

Task 1) Scale the Level meter 'ID30' in TIA in percentage (0-100%) and height (0-300 cm).

```
//Convert the range 0.0- 10.0 to Percentage 0-100%
#Scaled_Sensor := 10.0 * #SENSOR;

//Convert the range 0.0- 10.0 to Height range 0-300 cm
#Scaled_in_cm_real := #SENSOR * 30.0;
```

Task 2) Display the scaled values on the Panel

To display the scaled value on the Panel we have to convert Real into DINT. So for that following is the code:

```
#"Scaled_in cm" := REAL_TO_INT(#Scaled_in_cm_real);
```

But unfortunately the Panel Display was not working properly and has a software bug. However it will works perfectly if you use HMI in Siemens TIA

Task 3) Make an Auto FC using SCL code such that it fills the tank in such a way that:

If tank Level is < 40% = Fill Valve Opening 100%
If tank Level is > 40% and < 80% = Fill Valve Opening 50%
If tank Level is > 80% = Fill Valve Opening 0%
Use Toggle button to drain the level (at 50% flow rate)
Automatic ON/OFF CONTROL

```
//IF tank level is < 40% = Fill Valve Opening 100%
//IF tank level is > 40% AND < 80% = Fill Valve Opening 50%
//IF tank level is > 80% = Fill Valve Opening 0%

IF #"Level in Percentage" < 40.0 THEN
    #"Fill Valve" := 10.0;  // Statement section IF

ELSIF #"Level in Percentage" > 40.0 AND #"Level in Percentage" < 80.0   THEN
    #"Fill Valve" := 5.0; // Statement section ELSIF

ELSIF  #"Level in Percentage" > 80.0 THEN
    #"Fill Valve" := 0.0;// Statement section ELSE

END_IF;

//Use Toggle button TO drain the level (at 50% flow rate)
IF #"Drain Button" = 1 THEN
    #"Drain Valve" := 5.0;
END_IF;
```

Task 4) Make an arrangement to change the Set Point of Water Level using Potentiometer (0 – 100%). Write a code in SCL to achieve the set point using only Proportional Control.

Automation Proportional CONTROL

```
//Error
#error := #"Set Point" - #SENSOR;

//Proportional Control
#u := #error * #kpp;

//Output
#"Filling Valve" := #u;
#"Drain Valve" := - #u; // SP = 50% and Level is 100%
```

Task 5) Install a Potentiometer in Panel for Proportional Gain with your calibrated minimum and maximum limit.

```
// Input ID38 0-10.0 == 1 to 10
// Scaling Formulae
// Scaled Output = ((Output Max - Output Min)*(Input to be scaled - Input min.))/(Input max - Input min) + Output Min.
// In our case:
// Input Min = 0, Input Max = 10, Output min = 1,
// Output Max = 10 and Input to be scaled is Kp (Proportional gain from Pot)
#kpp := (((10.0 - 1.0)/(10.0 - 0.0))*(#Kp - 0.0))+ 1.0;
```